Tarea 4 y 5

Capitulo 4

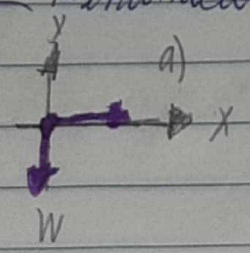
4.43

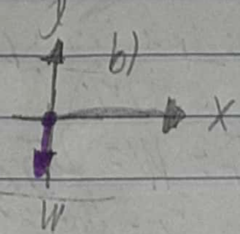
$\Sigma F_x = ma_x$
$T = m_1 a = (4)(2.5) = \underline{10N}$

$\Sigma F_x = ma_x \qquad F - T = m_2 a$
$F = T + m_2 a = 10 + (6)(2.5) = 10 + 15 = 25N$
$m = m_1 + m_2 = 10kg$

$\Sigma F_x = ma_x$
$F = ma = (10)(2.5) = \underline{25N}$

4.49 Dinámica de insecto

a) 



b) $V_{0y} = 0 \qquad V_y = 4m/s \qquad t = 1 \times 10^{-3} s \qquad V_y = V_{0y} + a_y t$

$a_y = \frac{V_y - V_{0y}}{t} = \frac{4 - 0}{1 \times 10^{-3}} = 4 \times 10^3 m/s^2 \qquad \Sigma F_y = ma_y \qquad n - w = ma$
$n = w + ma = m(g + a)$
$m(g+a) = (12.3 \times 10^{-6})(9.8 + 4 \times 10^3) = \underline{0.049N}$

c) $\frac{F}{W} = \frac{0.049}{(12.3 \times 10^{-6})(9.8)} = \underline{410w}$

4.53

a) La tensión en el cable debe ser igual al peso w, la fuerza en un punto del cable es cero

b) No hay tensión en la parte superior, la fuerza sobre el cable debe ser 0, en la parte inferior no hay tensión.

c) La tensión debe ser $W - \frac{W}{2} = \frac{W}{2}$ en el medio.

Capitulo 5

Capitulo 5

5.1

a) $\Sigma F_y = ma_y$. $T_r = W = 25N$

b) $T_c = 2T_r = 50N$

5.3

a) $\Sigma F_y = 0$ $T_{max} - m_{bic}\, g = 0$

$T_{max} = (101)(9.8) = 990N$

$T_{min} - m_b\, g = 0$

$T_{min} = (75)(9.8) = 735N$

Scanned with CamScanner

b) $T-(m+m_b)g=0$

$T=(19.5+75)9.8=\underline{926N}$

5.5

$T=0.75w \qquad \frac{W}{2}=\frac{3W}{4}\cos\theta \qquad \theta=\cos^{-1}\frac{2}{3}=\underline{48°}$

5.7

a) $\Sigma F_y=m\cancel{a}=0=T_A\,sen30+T_B\,sen45-W$

$T_A sen30+\left(\frac{T_A\cos30}{\cos45}\right)sen45=0$

$T_A\,sen30+T_A(\cos30\tan45)-W=0$

$T_A=\underline{0.732\,W}$

b) $\Sigma f_x=m\cancel{a}=0=T_B\cos45-T_A\cos30$

$T_B\cos45=T_A\cos30$

$T_B=\frac{T_A\cos30}{\cos45}=\underline{0.896W}$

5.9

a) $\Sigma F_x=0 \qquad F-W\,sen77=0 \qquad F=(186)(9.8)sen77$

$F=\underline{337\,N}$

b) $\Sigma F_y = 0$ $\quad n\cos 11 - W = 0$ $\quad n = \dfrac{W}{\cos 11}$

$\Sigma F_x = 0$ $\quad F - n\,sen\,11 = 0$

$F = \left(\dfrac{W}{\cos 11}\right) sen\,11 = W\tan 11 = \underline{343\,N}$

5.17 ¡Permanezca despierto!

a) $\Sigma F_y = ma$ $\quad F_T - W_r = ma$ $\quad a = 4g$ $\quad W_r = mg$

$F = m(5) = (2.25 \times 10^6)(5)(9.8) = \underline{1.10 \times 10^8\,N}$

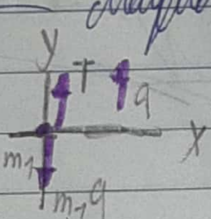
b) $\Sigma F_y = ma$ $\quad n - W = ma$ $\quad a = 4g$ $\quad m = \dfrac{W}{g}$

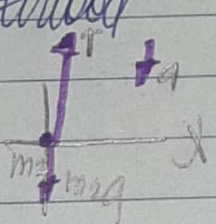
$n = W + \left(\dfrac{W}{g}\right)(4g) = \underline{5W}$

c) $V_0 = 0$ $\quad V = 337\,m/s$ $\quad a = 4 = 39.2$

$V = V_0 + at$ $\quad t = \dfrac{V - V_0}{a} = \dfrac{337}{39.2} = \underline{8.4\,s}$

5.15 Máquina de Atwood

a) 



b) ladrillos = $\Sigma F_y = ma_y$

$$T - m_1 g = m_1 a$$

contrapeso = $\Sigma F_y = ma_y$

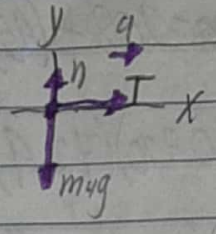
$$m_2 g - T = m_2 a$$

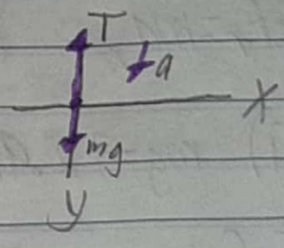
$$(m_2 - m_1)g = (m_1 + m_2)a$$

$$a = \left(\frac{m_1 - m_2}{m_1 + m_2}\right) g = \left(\frac{28 - 75}{75 + 28}\right) 9.8 = 2.96 \, m/s^2$$

c) $T - m_1 g = m_1 a$ $\quad T = m_1(a + g) = 75(2.96 + 9.8) = 791N$

5.17

a) a) 

b) 

b) $\Sigma F_x = ma_x$ $\quad T = (4kg)a$ $\quad a = \frac{T}{4} = \frac{10}{4} = 2.5 m/s^2$

c) $\Sigma F_y = ma_y$ $\quad mg - T = ma$ $\quad m = \frac{T}{g - a} = \frac{10}{9.8 - 2.5} = 1.37 kg$

d) $mg = (1.37)(9.8) = 13.4N$ es mayor la tensión de la cuerda) $T = 0.75mg$

5.25 $f_s = \mu_s n$ $\Sigma F_x = ma_x = 0$

$\Sigma F_y = ma_y$ $n = mg\cos\theta$

$\Sigma F_x = 0$ $mg\,sen\theta - \mu_s n = 0$

$mg\,sen\theta - \mu_s mg\cos\theta = 0$ $\tan\theta = \mu_s$ $\theta = 50°$

5.29

a) $max f_s = 313N$ $\mu_s = \frac{313}{441} = 0.710N$

$f_k = 208N$ $\mu_k = \frac{208}{441} = 0.472$

b) $\Sigma F_x = ma_x$ $F - f_k = ma$ $F = f_k + ma = 208 + (45)(1.10) = 258N$

c) i) $mg = 72.9N$ $F = max f_s = \mu_s n = (0.710)(72.9) = 51.8N$

ii) $F = f_k + ma$ $a = \frac{F - f_k}{m} = \frac{258 - (0.472)(72.9)}{45} = 4.97 m/s^2$

5.31 rampa $=$ $\tan\theta = \frac{2.5m}{4.75m} =$ $\theta = 27.76°$

$m_{tot} = 32 + 48 = 80Kg$

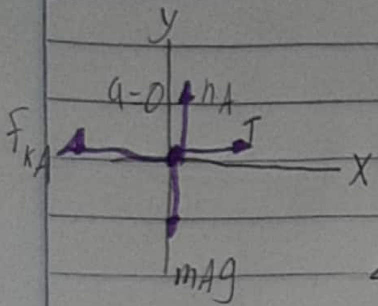
a) $\Sigma F_y = ma_y$ $n_{tot} = m_{tot} g\cos\theta$

$f_k = \mu_k m_{tot} g\cos\theta$ $\Sigma F_x = ma$ $f_k + T - m_{tot} g\sin\theta = 0$

$T = (\sin\theta - \mu_k\cos\theta) m_{tot} g = (\sin 27.76^\circ - 0.444\cos 27.76^\circ)(80)(9.8) = 57.1\ N$

b) $\Sigma F_x = ma_x$ $f_s = mg\sin\theta = (32)(9.8)\sin 27.76^\circ = 146\ N$

5.35



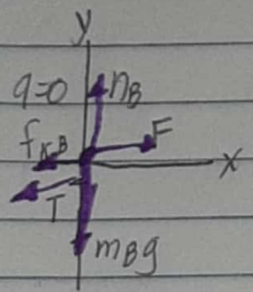
$\Sigma F_y = ma_y$ $n_A - m_A g = 0$

$n_A = m_A g$ $f_{kA} = \mu_k n_A = \mu_k m_A g$

$\Sigma F_x = ma_x$

$T - f_{kA} = 0$

$T = \mu_k m_A g$



$\Sigma F_y = ma_y$

$n_B - m_B g = 0$

$n_B = m_B g$

$f_{kB} = \mu_k n_B = \mu_k m_B g$

$\Sigma F_x = ma_x$

$F - T - f_{kB} = 0$

$F = T + \mu_k m_B g$

$F = \mu_k m_A g + \mu_k m_B g$ a) $F = \mu_k (m_A + m_B) g$

b) $T = \mu_k m_A g$